

Why These Losses?

A probabilistic refresher for Deep Learning

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ES 667 · Deep Learning · IIT Gandhinagar

Three familiar equations

You have already met these losses:

$$\mathcal{L}_{\text{reg}} = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2 \quad (\text{regression})$$

$$\mathcal{L}_{\text{class}} = -\frac{1}{n} \sum_i \log p_{\theta}(y_i \mid x_i) \quad (\text{classification})$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda \|\theta\|_2^2 \quad (\text{regularized})$$

Why exactly these three expressions? Today: they are not arbitrary.

Two probabilistic assumptions, two halves of the objective:



Minimal probability revision

A model should describe uncertainty

$$\hat{y} = f_{\theta}(x)$$

a single number

$$Y \mid X = x \sim p_{\theta}(\cdot \mid x)$$

a full distribution

- regression network → a **Gaussian mean**
- classifier → **categorical** probabilities
- language model → categorical distribution over **tokens**

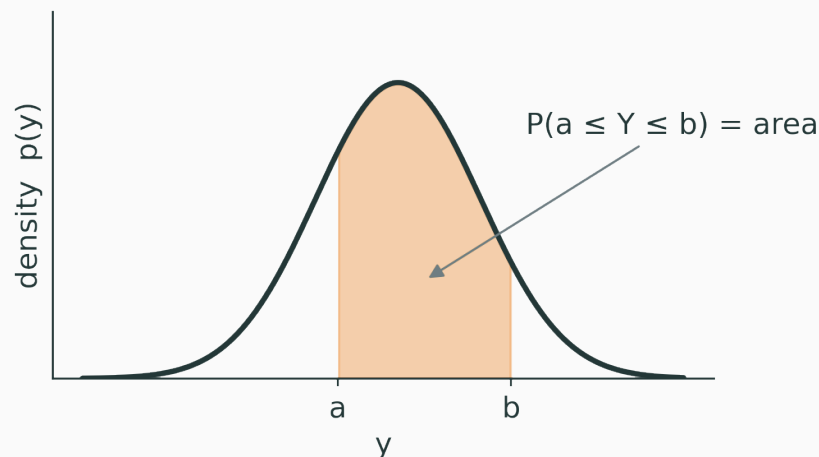
A neural network parameterizes a probability distribution.

Density is not probability

For a continuous Y , probability comes from **area**:

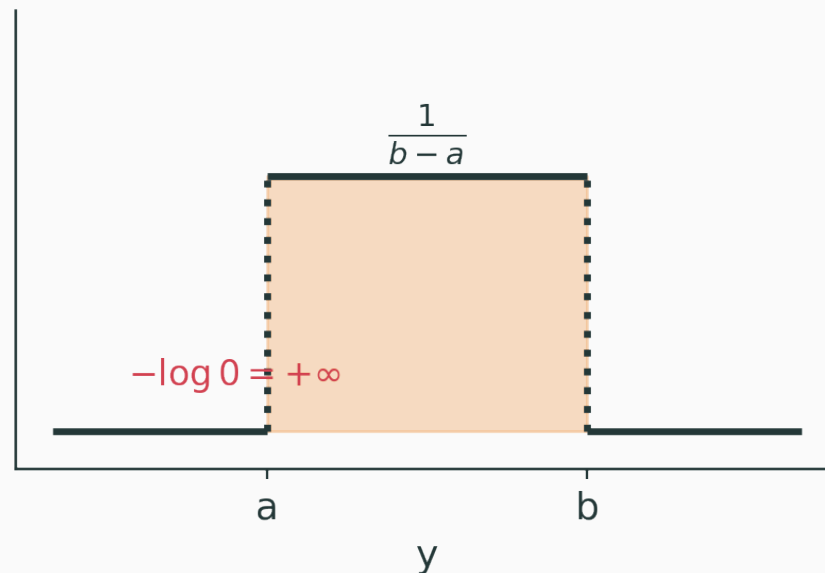
$$P(a \leq Y \leq b) = \int_a^b p(y) \, dy$$

$p(y) \neq P(Y = y)$ — a density can **exceed 1**; only *integrals* are probabilities.



$$Y \sim \mathcal{U}(a, b) \quad p(y) = \begin{cases} 1/(b-a) & a \leq y \leq b \\ 0 & \text{otherwise} \end{cases}$$

A likelihood can impose a **hard constraint**.



For a $\text{Uniform}(a, b)$ observation model, what is the NLL if the target lies outside $[a, b]$?

A. 0

B. A fixed finite penalty

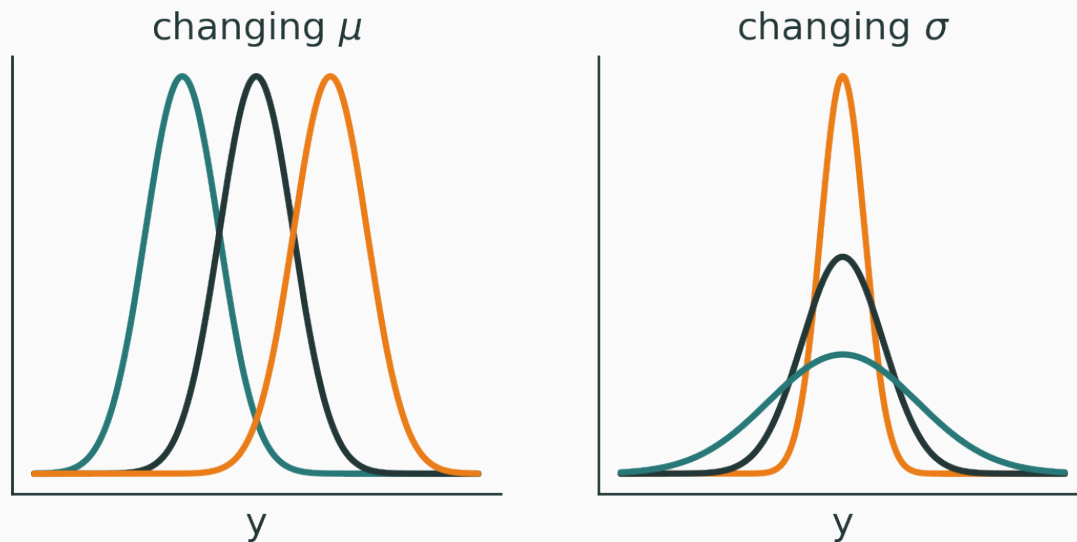
C. $-\log 0 = +\infty$

D. It depends only on σ

Correct: C — $-\log 0 = +\infty$

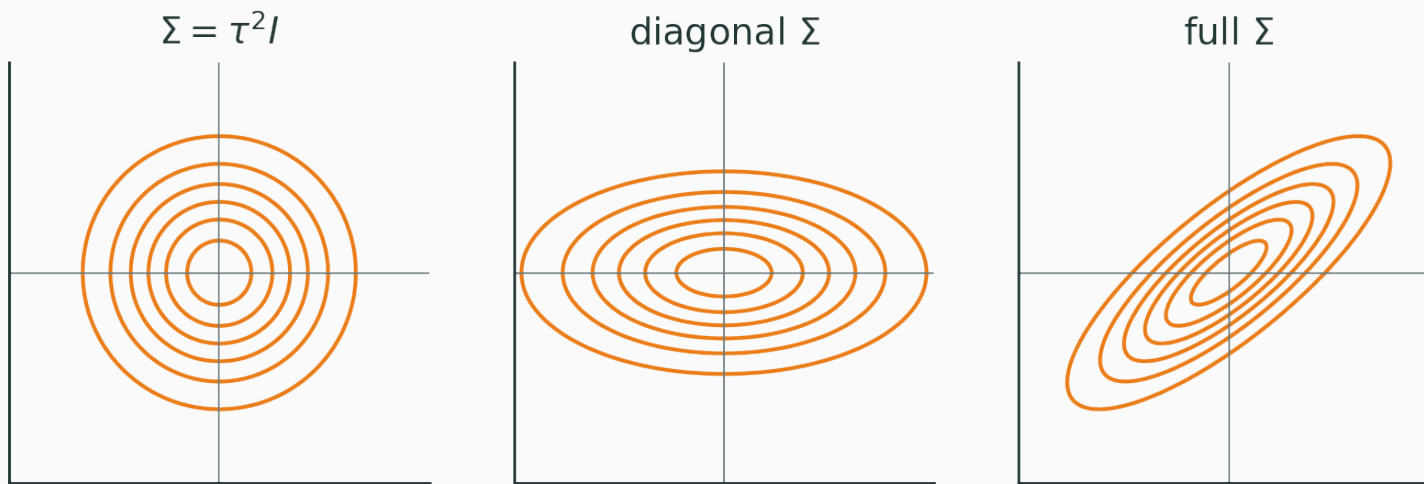
Why: The model assigns zero density outside its support, so such a target is impossible under the assumption.

$$Y \sim \mathcal{N}(\mu, \sigma^2) \quad p(y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y - \mu)^2}{2\sigma^2}\right)$$



Only two knobs: **location** μ and **scale** σ .

$$\theta \sim \mathcal{N}(\mu, \Sigma) \quad -\log p(\theta) = \frac{1}{2}(\theta - \mu)^\top \Sigma^{-1}(\theta - \mu) + C$$



Covariance says **which directions in parameter space are plausible.**

Independence turns products into sums

Conditionally independent observations:

$$p(\mathcal{D} \mid \theta) = \prod_{i=1}^n p_{\theta}(y_i \mid x_i)$$

Take logs — the product becomes a **sum**:

$$\log p(\mathcal{D} \mid \theta) = \sum_{i=1}^n \log p_{\theta}(y_i \mid x_i)$$

Why a loss is an **average over examples**, and why minibatches work: $\frac{1}{|B|} \sum_{i \in B} \ell_i$.

Bayes' rule → likelihood → loss

Turn the product rule $p(\mathcal{D}, \theta) = p(\mathcal{D} \mid \theta) p(\theta) = p(\theta \mid \mathcal{D}) p(\mathcal{D})$ around:

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta) p(\theta)}{p(\mathcal{D})}$$

$p(\theta \mid \mathcal{D})$	posterior — belief about θ <i>after</i> seeing data
$p(\mathcal{D} \mid \theta)$	likelihood — how well θ explains the data
$p(\theta)$	prior — belief about θ <i>before</i> data
$p(\mathcal{D})$	evidence — the normaliser $\int p(\mathcal{D} \mid \theta) p(\theta) \mathrm{d}\theta$

MLE (now): keep only the **likelihood**. **MAP (later):** likelihood $\times$ **prior** $\rightarrow$ regularisation.

**Maximum likelihood → training
losses**

Likelihood: one expression, two viewpoints

The one model $p_{\theta}(y \mid x)$, read two ways:

Fix θ , vary y

→ a **distribution** over outcomes.

Fix observed y , vary θ

→ a **likelihood** over parameters.

The likelihood is *not* “the probability that θ is correct.”

Ten flips come up $H, H, T, T, T, H, H, T, T, T$ — so $n_H = 4$, $n_T = 6$. Model each flip as $\text{Bernoulli}(\theta)$: $p(H) = \theta$, $p(T) = 1 - \theta$. The flips are i.i.d., so the **likelihood** of this exact sequence is

$$L(\theta) = \prod_i p(x_i \mid \theta) = \theta^{n_H} (1 - \theta)^{n_T} = \theta^4 (1 - \theta)^6$$

Different θ make the **observed** sequence more or less probable. Which θ makes it most probable?

Logs turn the product into a sum:

$$\mathcal{L}(\theta) = \log L(\theta) = n_H \log \theta + n_T \log(1 - \theta)$$

Set the derivative to zero:

$$\frac{d\mathcal{L}}{d\theta} = \frac{n_H}{\theta} - \frac{n_T}{1 - \theta} = 0$$

$$\Rightarrow \theta_{\text{MLE}} = \frac{n_H}{n_H + n_T} = \frac{4}{10} = 0.4$$

The intuitive “fraction of heads” *is* the maximum-likelihood estimate.

Maximum likelihood estimation

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} p(\mathcal{D} \mid \theta)$$

Interpretation. Choose the parameters under which the observed dataset would be **least surprising**.

From MLE to negative log-likelihood

$$\arg \max_{\theta} \prod_i p_{\theta}(y_i \mid x_i) = \arg \max_{\theta} \sum_i \log p_{\theta}(y_i \mid x_i)$$

$$= \arg \min_{\theta} \left(- \sum_i \log p_{\theta}(y_i \mid x_i) \right)$$

$$\ell_i(\theta) = -\log p_{\theta}(y_i \mid x_i)$$

NLL is empirical risk

ML notation — empirical risk:

$$\hat{R}(\theta) = \frac{1}{n} \sum_i \ell(y_i, f_\theta(x_i))$$

Probabilistic reading of the per-example loss:

$$\ell(y_i, f_\theta(x_i)) = -\log p_\theta(y_i \mid x_i)$$

Choosing a loss = choosing a probabilistic observation model.

**Regression — where does MSE
come from?**

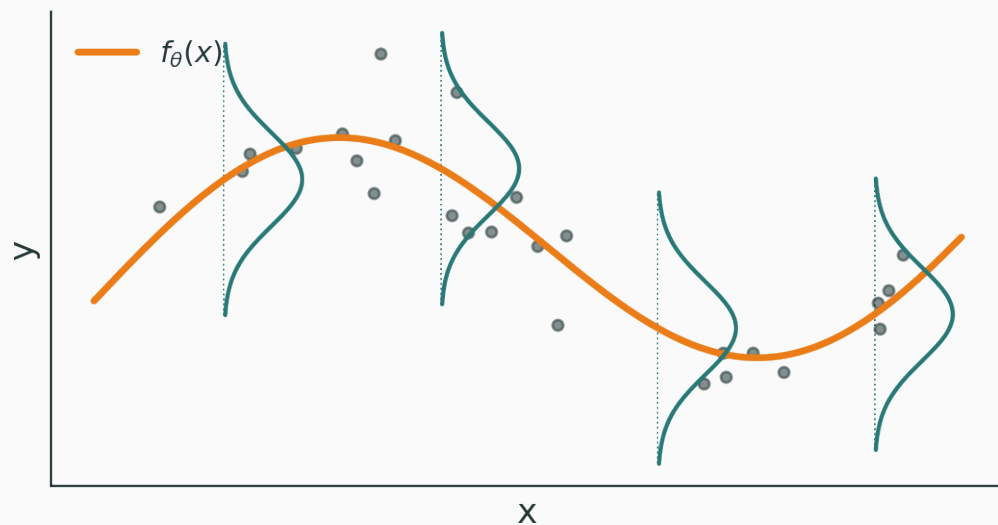
Regression as signal plus noise

$$y_i = f_{\theta}(x_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

Therefore the conditional output distribution is

$$Y_i \mid x_i, \theta \sim \mathcal{N}(f_{\theta}(x_i), \sigma^2)$$

What the assumption means visually



Gaussian noise lives in the **output direction**, conditional on x — a vertical bell at every input.

$$p_{\theta}(y_i \mid x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - f_{\theta}(x_i))^2}{2\sigma^2}\right)$$

Negative log of a single example:

$$-\log p_{\theta}(y_i \mid x_i) = \frac{(y_i - f_{\theta}(x_i))^2}{2\sigma^2} + C$$

MSE is Gaussian MLE

Sum over the dataset:

$$-\log p(\mathcal{D} \mid \theta) = \frac{1}{2\sigma^2} \sum_i (y_i - f_\theta(x_i))^2 + C$$

With σ fixed, the minimizer is

$$\hat{\theta}_{\text{MLE}} = \arg \min_{\theta} \sum_i (y_i - f_\theta(x_i))^2$$

MSE = Gaussian NLL, up to constants.

What role does σ^2 play?

★ OPTIONAL

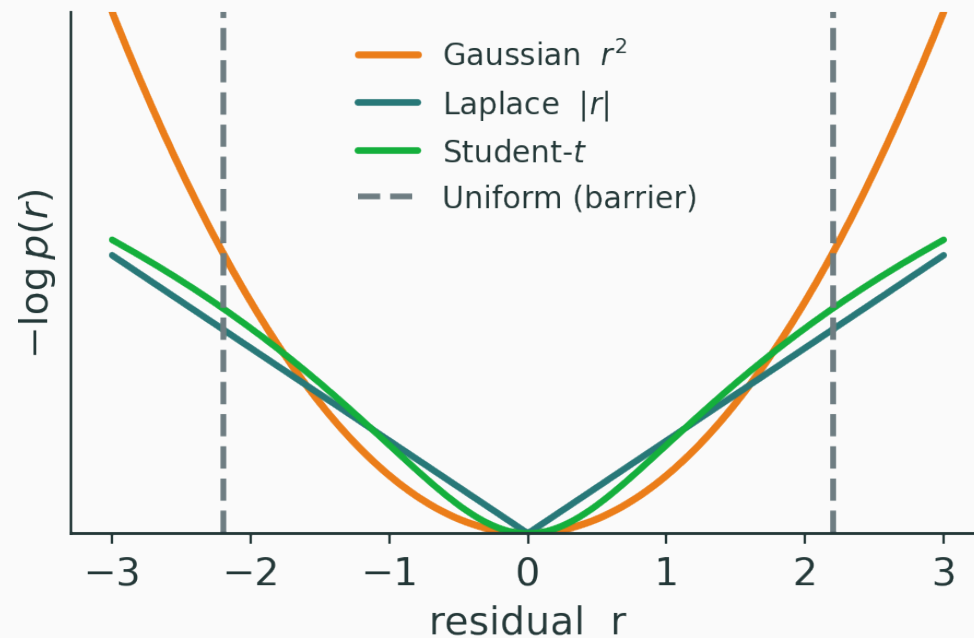
$$\ell_i = \frac{(y_i - \mu_i)^2}{2\sigma^2} + \frac{1}{2} \log \sigma^2 + C$$

- small σ : errors penalized **strongly**
- large σ : errors deemed **less surprising**
- a predicted σ_i needs the $\frac{1}{2} \log \sigma_i^2$ term — it stops the model claiming infinite uncertainty

Heteroscedastic regression: let the network output both $(\mu_i, \log \sigma_i^2) = f_\theta(x_i)$.

Noise model	NLL
Gaussian	r^2
Laplace	$ r $
Uniform	hard interval
Student- t	robust

$$r = y - \hat{y}$$



INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/likelihood-to-loss

likelihood-to-loss.html — three synchronized panels: the **noise density** $p(r)$, the **loss** $-\log p(r)$, and **data with a fitted line**.

Controls: Gaussian / Laplace / Uniform / Student- t · noise scale · slope & intercept · **add an outlier** · per-point contributions.

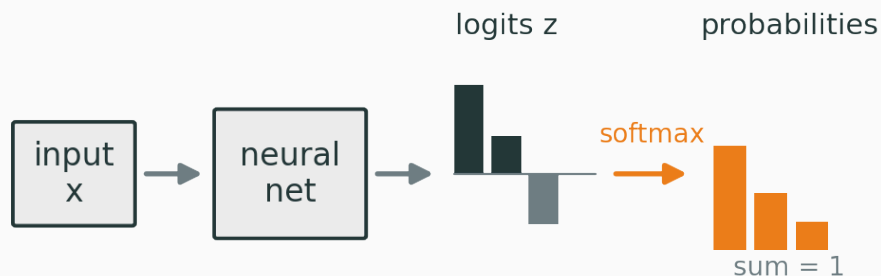
Ask: which model reacts most to an outlier? why does Uniform give an infinite barrier? which loss is robust?

Classification and cross-entropy

Classification should output probabilities

For K classes:

$$p_{\theta}(y = k \mid x) \geq 0, \quad \sum_{k=1}^K p_{\theta}(y = k \mid x) = 1$$



logits $z_k \in \mathbb{R} \rightarrow$ **probabilities** $p_k \in [0, 1]$.

Binary classification: Bernoulli model

$$Y \mid x \sim \text{Bernoulli}(p_\theta(x))$$

Compact one-line form:

$$p(y \mid x, \theta) = p^y (1 - p)^{1-y}$$

$$y = 1 \Rightarrow p(y) = p$$

$$y = 0 \Rightarrow p(y) = 1 - p$$

$$\begin{aligned} -\log p(y \mid x, \theta) &= -\log(p^y(1-p)^{1-y}) \\ &= -y \log p - (1-y) \log(1-p) \end{aligned}$$

Binary cross-entropy = Bernoulli NLL.

Check. True label $y = 1$: if $p = 0.8$ the loss is $-\log 0.8 \approx 0.22$; if $p = 0.2$ it is $-\log 0.2 \approx 1.61$.

Why use logits?

The network emits an unbounded score $z = f_{\theta}(x) \in \mathbb{R}$; the **sigmoid** maps it to a probability:

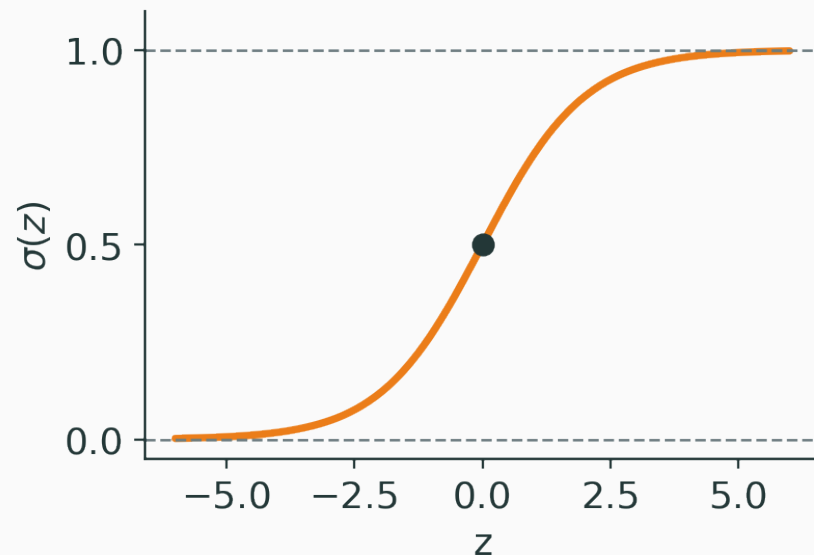
$$p = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$z \rightarrow -\infty : p \rightarrow 0$$

$$z = 0 : p = 0.5$$

$$z \rightarrow +\infty : p \rightarrow 1$$

In practice sigmoid + BCE are fused for numerical stability.



Predict each label with the sigmoid of a linear score: $p_i = \sigma(w^T x_i)$, and $Y_i \mid x_i \sim \text{Bernoulli}(p_i)$. Likelihood of the whole labelled dataset (i.i.d.):

$$L(w) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i}$$

Its negative log — the objective we minimise:

$$J(w) = - \sum_{i=1}^n [y_i \log \sigma(w^T x_i) + (1 - y_i) \log(1 - \sigma(w^T x_i))]$$

Exactly the per-example **BCE from two slides ago, summed over the data**. Now differentiate it in w .

One lemma — the sigmoid's own derivative:

$$\sigma'(z) = \sigma(z)(1 - \sigma(z))$$

Differentiate J through $\log \sigma$ and $\log(1 - \sigma)$; the $\sigma(1 - \sigma)$ cancels the chain factors and everything collapses to

$$\frac{\partial J}{\partial w_j} = \sum_{i=1}^n (\sigma(w^T x_i) - y_i) x_i^j$$

$$\Rightarrow \nabla_w J = \sum_{i=1}^n (\sigma(w^T x_i) - y_i) x_i = X^T (\sigma(Xw) - y)$$

The gradient is (prediction – target) $\times$ input — the same $p - y$ signal we will meet again for softmax.

Multi-class: categorical model

$$Y \mid x \sim \text{Categorical}(p_1, \dots, p_K)$$

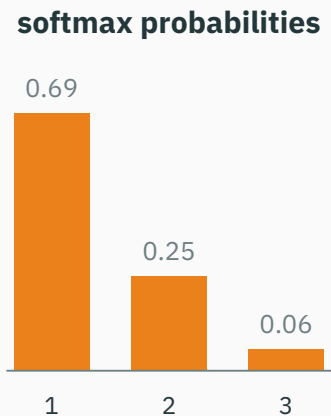
For a one-hot target y :

$$p(y \mid x, \theta) = \prod_{k=1}^K p_k^{y_k}$$

From logits to softmax

$$p_k = \frac{e^{z_k}}{\sum_j e^{z_j}}$$

Two guaranteed properties: $p_k > 0$ and $\sum_k p_k = 1$.



Shift invariance: $\text{softmax}(z) = \text{softmax}(z + c\mathbf{1})$.

Categorical NLL gives cross-entropy

D DERIVATION

$$-\log p(y \mid x, \theta) = -\log \prod_k p_k^{y_k} = -\sum_k y_k \log p_k$$

Because y is one-hot, only the true class survives:

$$\mathcal{L} = -\log p_{\text{true class}}$$

Confident mistakes are expensive

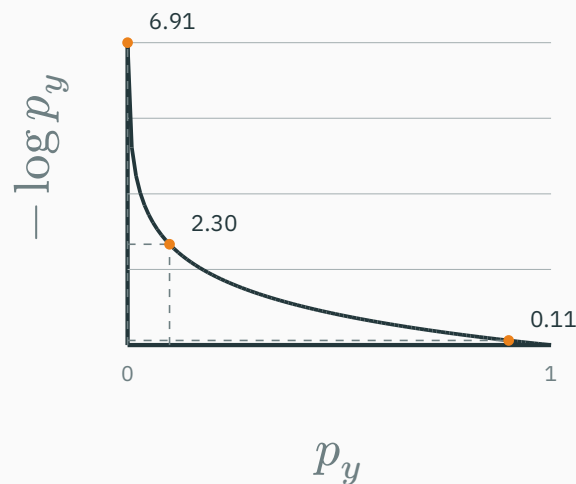
$$\ell(p_y) = -\log p_y$$

Read off the loss at three confidences for the **true** class:

$$p_y = 0.9 \Rightarrow \ell \approx 0.105$$

$$p_y = 0.1 \Rightarrow \ell \approx 2.303$$

$$p_y = 0.001 \Rightarrow \ell \approx 6.908$$



The negative log makes a confident wrong prediction a large loss.

Why does cross-entropy penalize a confidently wrong prediction strongly?

A. It makes all class probabilities equal

B. The true-class probability is near zero, so $-\log p_y$ is large

C. It adds an L_2 penalty to the logits

D. The gradient is always zero

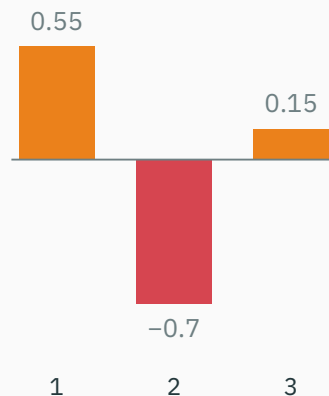
Correct: B — The true-class probability is near zero

Why: The negative log grows without bound as p_y approaches zero, matching the fact that the model declared the observed class nearly impossible.

For softmax followed by cross-entropy:

$$\frac{\partial \mathcal{L}}{\partial z_k} = p_k - y_k, \quad \nabla_z \mathcal{L} = p - y$$

$p - y$ (gradient on the logits)



For $p = (0.55, 0.30, 0.15)$, one-hot $y = (0, 1, 0)$: the signal $p - y$ is negative only on the true class — a preview of backprop.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/softmax-cross-entropy

softmax-cross-entropy.html — controls z_1, z_2, z_3 , the true class, and temperature T ; presets for **correct & confident / correct & unsure / wrong & unsure / wrong & confident**. Shows $z \rightarrow \text{softmax}(z/T) \rightarrow -\log p_y \rightarrow p - y$.

Ask students to predict the loss before it is revealed.

Why not MSE for classification?

★ OPTIONAL

MSE is not *impossible* — it corresponds to the **wrong observation model** for class indicators.

MSE assumes $Y_k = p_k + \varepsilon_k, \varepsilon_k \sim \mathcal{N}(0, \sigma^2)$

Reality is $Y \sim \text{Categorical}(p)$

Cross-entropy also gives **stronger gradients** on confident errors (where MSE's gradient vanishes).

Bayes' rule, directly in classification

Bayes' rule as a classifier

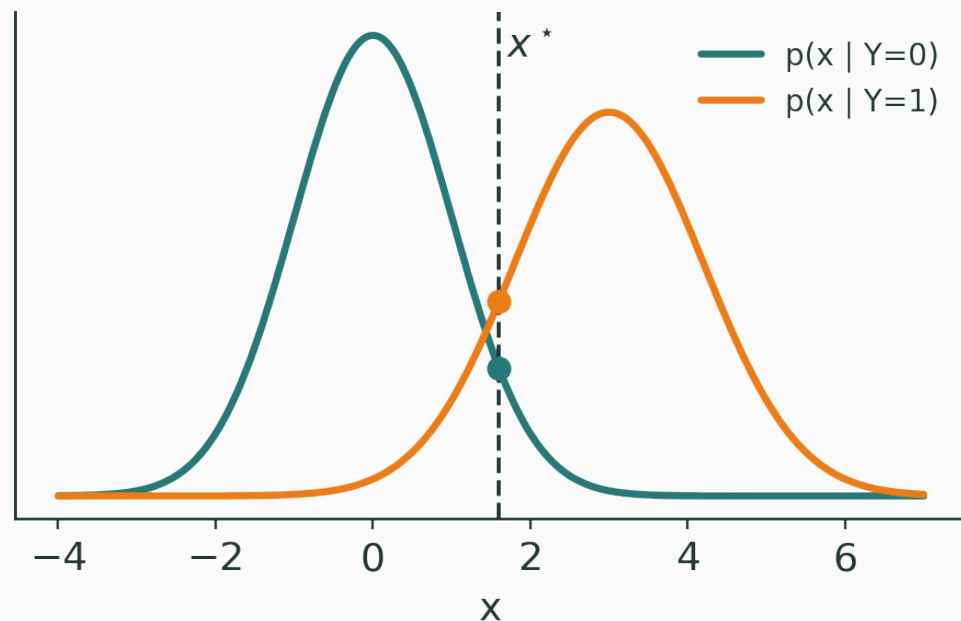
$$p(y = k \mid x) = \frac{p(x \mid y = k) p(y = k)}{p(x)}$$

Prediction ignores the constant denominator:

$$\hat{y} = \arg \max_k p(x \mid y = k) p(y = k)$$

- $p(y = k)$ — class **prior**
- $p(x \mid y = k)$ — class-conditional **likelihood**
- $p(y = k \mid x)$ — **posterior** class probability

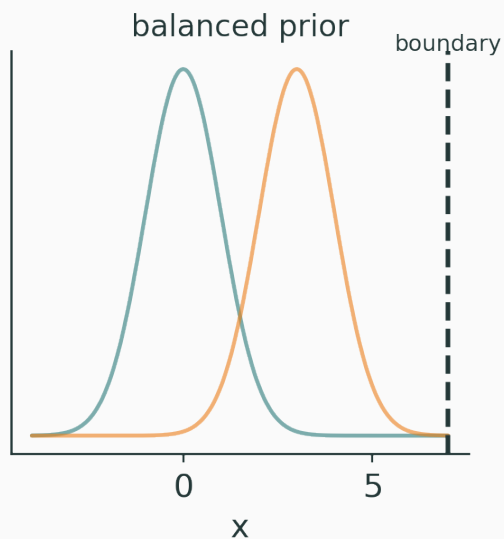
$$X \mid Y = 0 \sim \mathcal{N}(\mu_0, \sigma_0^2), \quad X \mid Y = 1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$$



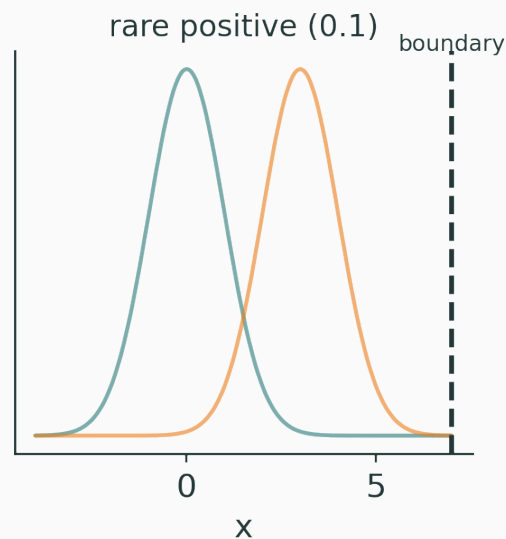
Two class-conditional densities, a test point x^* , and its likelihood under each class.

Priors move the decision boundary

Balanced $p(Y = 1) = 0.5$



Rare $p(Y = 1) = 0.1$



$$\text{posterior} \propto \text{likelihood} \times \text{prior}$$

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/bayes-classifier

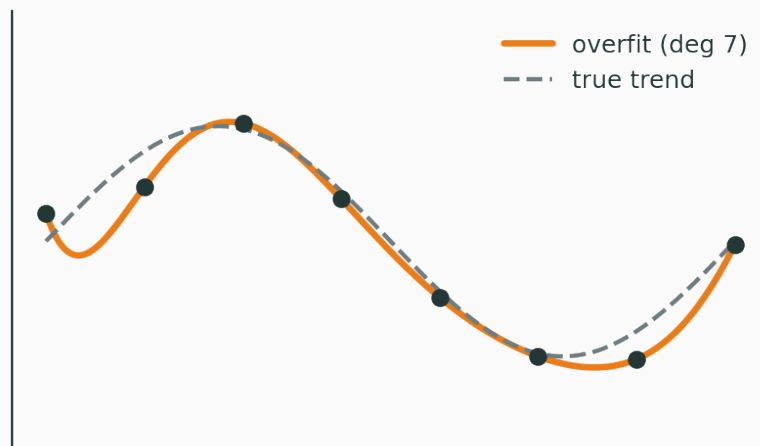
bayes-classifier.html — controls the means μ_0, μ_1 , the variances, the **prior** $p(Y = 1)$, and a test point x^* . Shows $p(x \mid Y = 0)$, $p(x \mid Y = 1)$, the posterior $p(Y = 1 \mid x)$, and the decision boundary together.

Try: balanced → drop the positive prior to 0.1 → why does the boundary move? then widen one variance.

Priors, MAP and regularization

The limitation of MLE

MLE asks only: *how well does θ explain the data?*



Many parameters fit a small dataset. **Which should we prefer?**

Put a prior over parameters

$$\theta \sim p(\theta)$$

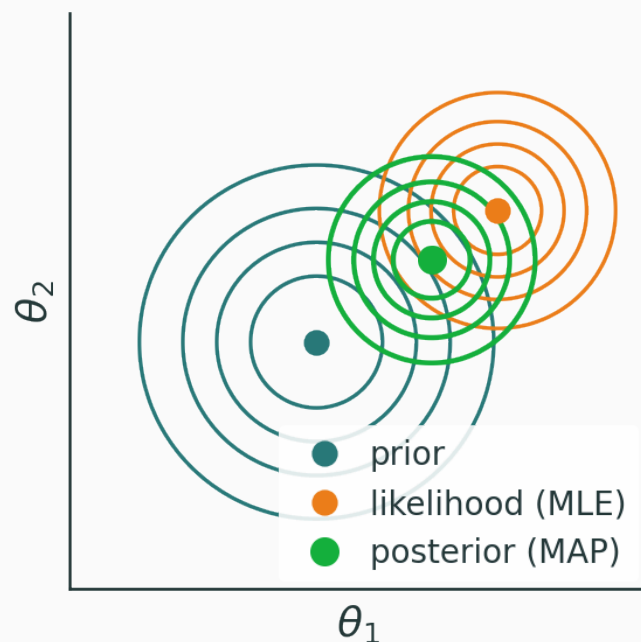
Before seeing this dataset, which parameter values are plausible?

- small weights · sparse weights · smooth functions · correlated parameters

A prior is an **inductive bias**.

Bayes' rule over parameters

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta) p(\theta)}{p(\mathcal{D})}$$



$p(\mathcal{D}) = \int p(\mathcal{D} \mid \theta) p(\theta) d\theta$ – the evidence (we stop here).

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta \mid \mathcal{D}) = \arg \max_{\theta} p(\mathcal{D} \mid \theta) p(\theta)$$

Take negative logs:

$$\hat{\theta}_{\text{MAP}} = \arg \min_{\theta} (-\log p(\mathcal{D} \mid \theta) - \log p(\theta))$$

data loss + regularization

Assume $\theta \sim \mathcal{N}(0, \tau^2 I)$. Then

$$-\log p(\theta) = \frac{1}{2\tau^2} \theta^\top \theta + C$$

$$\Rightarrow \mathcal{L}_{\text{MAP}} = -\log p(\mathcal{D} \mid \theta) + \lambda \|\theta\|_2^2, \quad \lambda \propto \frac{1}{\tau^2}$$

narrow prior $\Rightarrow$ **stronger** regularization; **broad** prior $\Rightarrow$ **weaker**.

A general Gaussian prior $\rightarrow$ structured penalty

★ OPTIONAL

For $\theta \sim \mathcal{N}(0, \Sigma)$ the penalty is $\frac{1}{2}\theta^\top \Sigma^{-1}\theta$.

- isotropic $\Sigma \rightarrow$ ordinary L_2
- unequal variances $\rightarrow$ penalize parameters **differently**
- correlations $\rightarrow$ penalize particular **directions**

$$p(\theta_j) = \frac{1}{2b} \exp\left(-|\theta_j| \frac{1}{b}\right) \Rightarrow -\log p(\theta) = \frac{1}{b} \sum_j |\theta_j| + C$$

Laplace prior $\Rightarrow L_1$

Heavier tails at zero $\rightarrow$ **sparsity** (the lasso geometry, if time permits).

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/map-regularization

map-regularization.html — a two-parameter linear model so contours are visible. Shows **likelihood**, **prior**, and **posterior** contours with the **MLE**, **MAP**, and zero points.

Controls: amount of data · noise · prior variance · Gaussian vs Laplace · prior correlation.

Ask: more data? narrower prior? why does MAP → MLE as data grows? a correlated prior?

Full Bayes and the deep-learning objective

MAP is still one parameter vector

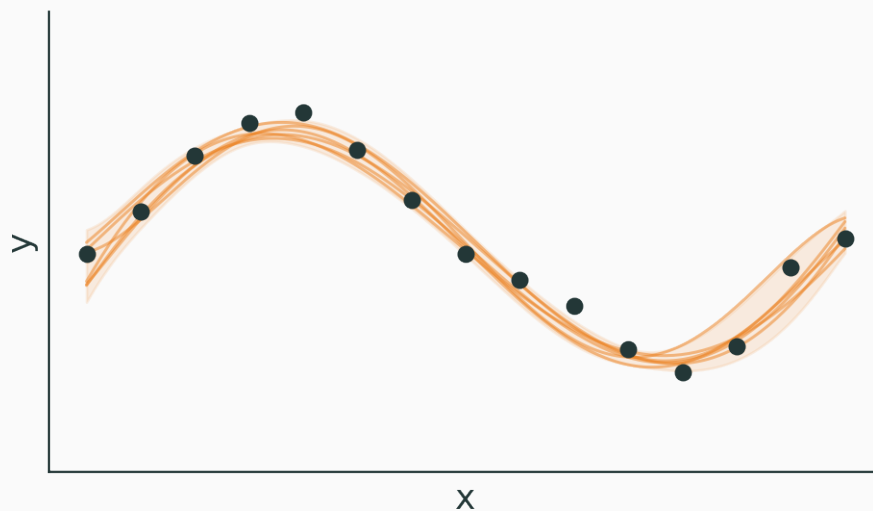
MLE $\hat{\theta}_{\text{MLE}}$ — one point

MAP $\hat{\theta}_{\text{MAP}}$ — one point

Full Bayes keeps the whole posterior $p(\theta \mid \mathcal{D})$ — a **distribution**, not a point.

Posterior predictive distribution

$$p(y^* \mid x^*, \mathcal{D}) = \int p(y^* \mid x^*, \theta) p(\theta \mid \mathcal{D}) d\theta$$



Bayesian prediction **averages over plausible models** — and reports uncertainty.

A modern network has millions–billions of parameters, so integrating $p(\theta \mid \mathcal{D})$ is hard. Common practical choices:

- MLE · MAP
- deep **ensembles**
- **dropout**-based approximations
- **variational** inference

The standard supervised DL objective

$$\min_{\theta} \underbrace{\frac{1}{|B|} \sum_{i \in B} -\log p_{\theta}(y_i \mid x_i)}_{\text{likelihood / data fit}} + \underbrace{\lambda \|\theta\|_2^2}_{\text{prior / regularization}}$$

probabilistic model + neural network + gradient optimization

DL doesn't discard probabilistic ML — it **scales** it with expressive functions and gradient descent.

Gaussian likelihood

⇒ **MSE**

Categorical likelihood

⇒ **cross-entropy**

Gaussian prior

⇒ L_2

Laplace prior

⇒ L_1

MLE + prior

⇒ **MAP**

Final mental model

See a **loss** → ask “*what likelihood produced it?*”

See a **regularizer** → ask “*what prior produced it?*”

See a **prediction** → ask “*point estimate or posterior?*”

Next: computation graphs, gradients, and backpropagation.